

CS564 – Machine Learning

Assignment 2: Bayesian Classifier

Date: Oct 10, 2025 **Deadline:** Oct 17, 2025, 11:59 PM IST

Instructions

1. Code must be properly **indented, commented**, and well-structured.
2. Marks will be deducted in case of **plagiarism**.
3. All required files must be compressed and named as:

`rollno1_rollno2_rollno3.zip`

e.g., `2021cs11_2021cs03_2021cs05.zip`.

4. Upload the assignment to: <https://forms.gle/FvMDWxtc2mHKiGG78>.

Problem Statement

You are required to build a **spam filtering system** using two variants of the Naive Bayes classifier:

- **Multinomial Naive Bayes**
- **Multivariate Bernoulli Naive Bayes**

Download the **SMS Spam Collection** dataset <https://archive.ics.uci.edu/ml/machine-learning-databases/00228/smsspamcollection.zip>, which contains:

- Total messages: 5,574
- Ham (legitimate): 4,827 (86.6%)
- Spam: 747 (13.4%)

Each line contains a label and a raw text:

`<label> <raw_text>`

Example:

ham What you doing? how are you?

spam URGENT! Your Mobile No 07808726822 was awarded a £2,000 Bonus!

Implementation Requirements

1. Text Preprocessing

- Tokenization, lowercasing
- Stopword removal
- Vocabulary construction

2. Model Implementation

- Implement both **Multinomial** and **Multivariate Bernoulli** Naive Bayes **from scratch**.
- Use **add-one (Laplace) smoothing**.
- Use log-probabilities to avoid numerical underflow.

3. Training & Testing

- Perform **5-fold cross-validation** with 90% in train set.
- Train the model on each fold and evaluate on the held-out test set.

4. Evaluation Metrics

- Accuracy
- Precision, Recall, F1-score
- Confusion Matrix
- Macro and Micro F1-scores
- Plot Precision-Recall and ROC curves.

Expected Output

- Confusion matrix for each fold and final averaged results.
- Accuracy, Precision, Recall, and F1-score for both models.
- A comparison table for **Multinomial** vs. **Multivariate** Naive Bayes.
- Plot ROC curves.
- All the outputs must be clearly visible in the submitted notebook.

Restrictions

- No pre-built Naive Bayes libraries are allowed (e.g., `scikit-learn`).
- Only standard Python libraries (`math`, `re`, `collections`, `matplotlib`, `numpy`) are permitted.
- Tokenization and stopword removal can use libraries such as NLTK or `spaCy`, but classification must be implemented manually.